Kishan Paschapur

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RESEARCH

A Controlled Analysis of Physics-Informed Augmentation for Powder X-Ray Diffraction Classification
K. Paschapur, P. Kulkarni (Stanford University). Under review, IEEE ONCON 2026. github.com/Kishan-No1/pxrd-physics-augmentation

- Built a parallel, resumable pipeline that simulates powder X-ray diffraction patterns from the Crystallography Open Database (~490K structures) and rasterizes them for a Swin-T vision transformer.
- Reproduced the published SIMPOD benchmark on the authors' split (82.92% vs. 82.69% reported synthetic top-5 accuracy).
- Designed physics-informed augmentation (peak broadening, background, counting noise, zero shift, texture) that raised top-5 accuracy on 20 real measured patterns from 30% to 55%, with gains in all three repeated runs.
- Ran a leave-one-component-out ablation identifying broadening and background as the main contributors; reported Wilson intervals and exact McNemar tests.
- Found and documented a fold-allocation flaw that silently removed the four most common space groups from training.

MIT Beaver Works Summer Institute (BWSI) · Serious Games & AI (SGAI) track *Summer 2026*

- Designed a serious game that measures human trust in AI decision-making in medical and combat scenarios.

TECHNICAL EXPERIENCE

Purpose Tech AI · Computer Science & Front-End Intern *Apr.–Jun. 2024*

- Engineered and tested AI prompts with Vercel AI, resolving backend logic issues and improving application reliability.
- Redesigned front-end UX and automated email templates to improve user engagement.

SKILLS

- **Machine learning:** PyTorch, torchvision (Swin-T), scikit-learn, NumPy, pandas, SciPy
- **Compute:** cloud GPU training on NVIDIA H100, B200 and RTX 4090; Apple MPS; bfloat16 mixed precision
- **Scientific computing:** Dans_Diffraction, gemmi (CIF parsing), statistical evaluation (McNemar tests, Wilson intervals)
- **Languages:** Python, Java

COURSEWORK

- Stanford CS229 Machine Learning (self-study, Dec. 2025–present)
- Harvard CS50 Introduction to Machine Learning with Python (self-study, Dec. 2024–Jan. 2025)
- Introduction to Programming in Python, De Anza College (Summer 2025) · AP Computer Science A (UC Scout)

TEACHING & LEADERSHIP

BAY Python Academy · Founder, Head of Curriculum *Jun.–Aug. 2024*

- Founded a local Python academy; designed and taught an 8-week course with a capstone project for 20+ middle schoolers.